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Dogs can Generalize from Cotton Training Aids to Fecal Matter in Chronic Wasting Disease Detection

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ABSTRACT: Chronic wasting disease (CWD) is a prion-associated transmissible spongiform encephalopathy (TSE) that causes fatal neurodegeneration in cervids. One promising method for CWD surveillance is the use of detection dogs trained to differentiate between fecal matter from CWD-positive (infected) and CWD-negative (noninfected) deer. However, using actual fecal matter during the training process confers a biohazard risk. Chronic wasting disease detection dogs trained on fecal matter can differentiate between CWD-positive and CWD-negative training aids, including cotton aids, which are inexpensive and simple to use. However, as most dogs in an applied setting would instead be trained on *aids* and tasked with finding actual fecal matter in the field, this study examined 1) the extent to which dogs initially trained on CWD-incubated cotton can discriminate between fecal samples from CWD-positive and CWD-negative deer; and 2) the impact of substrate form (cotton ball or cotton roll) on the dogs' detection accuracy. Results show that dogs ($n=5$) maintained an accuracy of $>80\%$ during training while discriminating between cotton ball training aids that had been incubated (noncontact) with deer fecal samples confirmed as either CWD-positive or CWD-negative. When tested with cotton ball training aids incubated with novel samples, dogs ($n=4$) showed 73% sensitivity and 77% specificity. When presented with deer fecal samples, dogs showed a sensitivity of 50% and a specificity of 90%. The type of cotton (ball or roll) did not significantly influence performance. In sum, cotton aids can be used to train dogs to detect the CWD odor profile effectively. However, an initial reduction in sensitivity suggests that integrating exposure to target fecal matter may enhance detection performance.

Key words: Biodetection, canine, chronic wasting disease, detection, disease, discrimination, dog, generalization, olfaction.

INTRODUCTION

Chronic wasting disease (CWD) is a transmissible spongiform encephalopathy (TSE) associated with prions, that leads to fatal neurodegeneration in cervid species, including white-tailed deer (*Odocoileus virginianus*), mule deer (*Odocoileus hemionus*), and elk (*Cervus canadensis*; Edmunds et al. 2016). Chronic wasting disease has been documented in free-ranging cervid populations across 35 US states (National Wildlife Health Center [NWHC] 2024). The impact of CWD can reverberate through ecosystems if cervid populations are significantly affected, whether because of the disease itself, preventative measures such as culling, or restrictions on hunting and farming (Rivera et al. 2019; Bhattarai et al. 2024). CWD is highly infectious

and is characterized by variable incubation periods before clinical signs (e.g., locomotive dysfunction, weight loss, ataxia) appear (Williams 2005). Both symptomatic and presymptomatic cervids can transmit the infective prions through biological excretions (Haley and Hoover 2015). Presently, CWD is always fatal, and there are no vaccines or therapies available. Although CWD has a low prevalence in most regions, there are localized pockets of infection that exceed 5%, and in endemic areas prevalence can reach over 85% (Osnas et al. 2009; Walsh 2012; Belsare et al. 2020; Bartz et al. 2024). Modeling suggests that detection at $<1\%$ prevalence in a new area is necessary for best management practices (Benestad et al. 2016). This underscores the critical importance of early detection methods for controlling the spread of CWD. Current

surveillance efforts rely heavily on hunter and landowner participation, with postmortem testing, or testing of the rectal anal mucosal associated lymphoid tissue, which requires handling individual deer to collect biopsies (US Department of Agriculture–Animal and Plant Health Inspection Service [USDA-APHIS], 2021).

Detection dogs represent a promising method to enhance early detection to curb the spread of CWD. Trained dogs have the potential to detect CWD-infected cervids via their excretions, most feasibly, their fecal matter. This type of antemortem, noninvasive surveillance would be particularly valuable, as many cervids do not show clinical signs typical of CWD for months to years after infection, making detection from visually tracking the animals ineffective. Furthermore, dogs have been shown to work effectively as scat (feces) detection dogs in the field, outperforming human-only teams in several previous studies (Long et al. 2011; Mathews et al. 2013; Orkin et al. 2016; Bennett et al. 2020).

Detection dogs for CWD would be tasked with two levels of discrimination: First, they need to discriminate between the target cervid species' fecal matter and other species' fecal matter; second, they must discriminate whether that fecal matter came from a cervid with CWD or a cervid not infected by CWD. The premise underpinning this concept is that when the cervid is infected by CWD, the volatile organic compound (VOC) profile in their fecal matter changes in ways that are detectable to a trained dog's nose. A recent study demonstrated that dogs could discriminate between the fecal matter of CWD-positive and CWD-negative deer both in laboratory and applied settings (Mallikarjun et al. 2023b).

The prions causing CWD are environmentally transmissible via cervid excreta. Prions remain infectious in the environment for at least 2 yr (Miller et al. 2004). CWD prions can bind to soil particles, and some evidence suggests that the soil particles enhance infectivity (Johnson et al. 2006, 2007; Pritzkow et al. 2015; Carlson et al. 2023). Furthermore, CWD prions show resistance to natural deactivation mechanisms

such as freeze–thaw cycles, extracellular enzymes from bacteria and fungi, and digestion by macrofauna in soil (Johnson et al. 2006). Because of a current gap in information underlying specific protocols, cleaning and disinfection guidelines for removing prions from soil are based on scrapie (a prion disease of sheep) and are limited to the removal of waste and infectious material, disposal of contaminated soil, and restriction of animals entering infected environments (Alarcon et al. 2021). Most published decontamination plans do not consider contamination in wild settings that are common in free-roaming cervid populations and of concern when considering CWD. Furthermore, handling CWD fecal samples has a high likelihood of environmental recontamination (Gough et al. 2019; Alarcon et al. 2021). As such, training detection dogs with fecal samples from CWD-positive cervids while monitoring containment and proper handling may not be feasible for most dog handlers. A potential solution is the use of training aids that represent the target and control odors and are safe to handle and place in the environment. However, it is necessary to establish first whether such a training aid successfully represents the target and control odors. Because operational dogs would be tasked with finding actual fecal matter (rather than the training aid), it would also be critical to assess whether dogs trained initially on training aids would generalize to the targets they would find in an applied situation, in this case, cervid fecal matter.

In a recent publication by Mallikarjun et al. (2024), cotton balls incubated with deer fecal samples at 21 C for 24 h captured the VOC profile of CWD-positive and CWD-negative samples, and dogs were able to categorize these samples with statistically significant accuracy. Further, dogs' performances were not significantly different when using inexpensive cotton ball aids as compared to other more expensive aid substrates, such as Getxent tubes (Getxent 2023) or polydimethylsiloxane polymer (Mallikarjun et al. 2024). Having established that cotton is a viable training aid to capture the odor profile of fecal samples from CWD-positive and CWD-negative deer,

TABLE 1. Demographic details of the five detection dogs trained at the Penn Vet Working Dog Center (PVWDC, Philadelphia, Pennsylvania, USA) to detect fecal matter from chronic wasting disease–infected white-tailed deer (*Odocoileus virginianus*). Each dog’s name, sex, age (y), reproductive status, breed, prior detection odor experience, and ownership are provided.

Dog name	Sex ^a	Age (y)	Reproductive status	Breed ^b	Odors learned prior to this study	Owner
Helen	F	4	Spayed	LR	UDC, ^c ovarian cancer	PVWDC
Vauk	F	1	Spayed	LR	UDC	PVWDC
Lucy	F	5	Spayed	DS	UDC, ovarian cancer	PVWDC
Anya	F	2	Spayed	LR × GR	UDC	Privately owned
Xoxo	F	4	Intact	BM	UDC, sinonasal inverted papilloma	Privately owned

^aF = female.

^bLR = labrador retriever; DS = Dutch shepherd; GR = golden retriever; BM = Belgian malinois.

^cUDC = Universal Detection Calibrant.

attention needs to be turned to the application of aids in the training process for CWD detection dogs. For example, the type of cotton substrate may be important when used in the field. Cotton balls and rolls are popular choices amongst working dog trainers; the denser cotton rolls are less likely to be disrupted by wind, and therefore may be more applicable in the field.

Considering these factors, this study examined two research objectives, the first being to determine the extent to which dogs initially trained on CWD-incubated cotton can discriminate between fecal samples from CWD-positive and CWD-negative deer. Given the results from prior studies, we hypothesized that dogs trained exclusively on the CWD training aid (in this case, cotton incubated with deer feces) would be able to discriminate between actual fecal samples taken from confirmed CWD-positive and CWD-negative deer; thus CWD training aids could be incorporated into a CWD detection training program. Second, we examined how substrate form (cotton ball or cotton roll) might affect the dogs’ detection accuracy.

METHODS

Animals

Five dogs participated in this study (see Table 1). The handlers of the privately owned dogs ($n=2$)

provided informed consent prior to their dog’s involvement in this study.

Samples and training aids

Samples from CWD-positive and CWD-negative deer were procured by the Wildlife Futures Program (WFP), as described (Mallikarjun et al. 2024). Deer were tested via immunohistochemistry on either the obex (brain) or large retroperitoneal lymph node (RPLN). Fecal samples were collected across the US. Testing for CWD was performed at the National Veterinary Services Laboratory, Ames, Iowa, US. Fecal samples for testing were selected randomly by WFP.

Cotton balls and rolls were incubated at 21 C for 24 h with randomly selected fecal samples from CWD-positive and CWD-negative deer by WFP, based on data from Mallikarjun et al. (2024). For training, cotton balls incubated with unique CWD-positive feces ($n=54$) and cotton balls incubated with unique CWD-negative feces ($n=79$) were used. Cotton rolls were not used in training. Training samples were used a maximum of four times per dog. If a sample was shown to the dog in a repeated session beyond the first presentation, it was considered “familiar.” At test, only the dogs’ responses to “novel” (e.g., never been encountered during training) samples were included for analysis.

Training

All sessions were recorded using a Panasonic HC-V380 video camera. Dogs were originally trained to detect Universal Detector Calibrant (UDC; Furton and Beltz 2017), a training odor used to teach the dogs scent wheel search mechanics

and to provide a trained final response (e.g., a sit or stand-stare) on the target sample. Dogs were additionally trained to perform a “blank wheel behavior,” whereby they searched each port of the scent wheel and sat on a table outside of the search area to indicate that no target odor was present. Samples were presented in a pseudo-randomized order within an eight-port scent wheel. The target sample could not appear in the same port for more than three trials in a row, and the dogs were not presented with more than two blank trials in a row. Control odors were included during training, which were a selection of distractor items that could co-occur with sample preparation, including nitrile gloves, nonincubated cotton balls or rolls, and isopropanol alcohol on paper towels. Dogs were trained to differentiate cotton aids incubated with CWD-positive and CWD-negative fecal samples from white-tailed deer. Prior to test, dogs achieved 80% sensitivity and 80% specificity across three consecutive training sessions. These training sessions were structured like the test sessions such that performance at test would accurately reflect dogs’ abilities and not reductions due to changes in the test environment.

Testing

Testing was conducted using a double-blind Bluetooth light system. Experimenters watched the sessions on a video feed and signaled to the handler when the dog had completed a 2-s stand-stare behavior (their trained final response) on the correct sample by remotely switching the colors of the Bluetooth lights to green. If it was a blank trial, the remote experimenters would switch the colors of the Bluetooth lights to green once the dog performed their blank wheel behavior. The lights were only visible to the handler on the opposite side of a wall divider, so the dogs could not see the lights. Once the light switched to green, the handler clicked (secondary reinforcer) and rewarded the dog.

Dogs were given two passes per trial, meaning that if the dog initially passed the positive sample and did not alert on any other sample in the lineup, the handler sent them on the same trial for a second time. If they were still unable to perform a correct final response, the experimenters moved to the next trial. However, only the dogs’ initial response to each odor on the first pass was considered in the analyses.

Dogs were presented with six sets of test samples, with each set presented in its own testing session.

The first three sets were to test their ability to classify novel cotton ball samples as indicative of being incubated with CWD-negative or CWD-positive fecal matter. The fourth test set was on cotton “rolls” instead of cotton balls to test whether cotton type impacted performance; this set, like all other cotton sets, contained a subset of novel samples never seen by the dogs. The fifth and sixth sets tested their ability to classify actual fecal samples from CWD-negative and CWD-positive white-tailed deer despite never encountering this true material previously.

Each test session consisted of a combination of training, blank, and probe trials. The order of the blank and probe trials varied across dogs to account for order effects. One to two “training” trials were shown to the dog using previously seen samples from training. Zero to two “blanks” (containing only negative samples) were shown to the dog. In these blank trials, some of the negatives were novel, and some were previously encountered. Lastly, 2–3 probe (test) trials were shown to the dog; these always contained one novel positive and at least one novel negative, with some inclusion of additional familiar negatives in specific circumstances due to limited availability of novel samples from WFP (see Table 2). Importantly, the dogs’ accuracy at test was only calculated on their response to novel samples.

Sensitivity and specificity were calculated for each test on dogs’ responses to novel samples. Sensitivity refers to the proportion of positive samples on which the dogs correctly alerted on their first encounters with the samples. Specificity refers to the proportion of negative samples the dogs correctly passed on their first encounters with the samples.

Video coding

Videos were obtained for each testing session. These data were coded using BORIS video coding software (Friard and Gamba 2016). Coders were unaware of the location of the positive and negative samples. Dogs’ duration at port was calculated as the time at which their nose was within 10 cm (4 in.) of the port. The dogs’ trained final alert was a stand-and-stare alert for at least 2 s; this coding verified that the alerts were at least 2 s long, and identified instances of false alerts, in which the dogs performed an alert at a negative or control sample.

TABLE 2. Test structure and trial composition for detection dogs trained to identify fecal matter from chronic wasting disease (CWD)-infected white-tailed deer (*Odocoileus virginianus*). Each dog ($n=5$) was trained at the Penn Vet Working Dog Center (Philadelphia, Pennsylvania, USA) using cotton training aids (cotton balls incubated with fecal samples from CWD-positive and CWD-negative deer). The table describes the structure of each testing session, including the test medium, total number of trials per session, number of training trials with previously encountered samples, number of blank trials, and number of probe test trials with novel positive or negative samples. Dogs were evaluated using an eight-port scent wheel under double-blind conditions to assess their sensitivity and specificity to novel samples.

Test	Medium	No. trials total	No. training trials	No. blanks (PS ^a included)	No. blanks (NS ^b only)	No. test trials (PS included)	No. test trials (NS only)
1	Cotton ball	6	1	2	0	3	0
2	Cotton ball	6	1	2	0	3	0
3	Cotton ball	6	1	2	0	3	0
4	Cotton roll	6	2	1	0	3	0
5	Fecal matter	4	2	0	0	0	2
6	Fecal matter	6	1	1	1	3	0

^a PS = previously seen samples.

^b NS = novel samples.

Analyses

All statistical analyses were carried out in R version 4.2.0 (R Core Team 2024). An interrater analysis was run to determine coder reliability for four randomly chosen test videos; duration of time spent at port was compared between coders. For the model, whether or not dogs correctly performed their trained final alert on sample was the outcome variable. If dogs performed their trained final alert on a CWD-positive sample, that indicated a correct alert; if dogs performed a trained final alert on a CWD-negative sample, that indicated a false alert. If they did not perform their alert on a CWD-positive sample, that was an incorrect pass of a positive sample; if they did not perform their alert on a CWD-negative sample, that was a correct pass of a negative sample. Analyses were conducted only on dogs’ responses to novel samples not used in training.

We used a binomial mixed-effects regression model to examine the effect of CWD status (CWD positive or negative) and sample medium (cotton ball, cotton roll, or fecal matter) on the dogs’ trained final responses. The model included Dog and Test Number as a random intercepts, as individual dogs can have variability in their baseline alert rates, and the test order or impacts of particular testing days may have impacted dogs’ test-by-test performance. Post hoc tests with a Tukey adjustment were used to explore further the impact of each CWD sample medium and CWD status on dogs’ alert behavior.

These tests were done using the emmeans function within the emmeans package in R (Lenth 2025).

RESULTS

Interrater reliability of coding

The video-coding interrater analysis used a single-rating, agreement, one-way mixed-effect model, which showed that the intraclass coefficient was 0.962, with a 95% confidence interval of 0.951–0.97. This is considered excellent reliability (Koo and Li 2016).

Model 1: Sample status differentiation and impact of sample medium

There was a significant main effect of Sample Status such that dogs performed their trained final response more often on CWD-positive samples than CWD-negative samples, $\beta=2.47$, $z=5.08$, $P=3.79 \times 10^{-7}$. There was no main effect of Sample Medium (cotton ball, cotton roll, or fecal matter). The model found no difference in overall alerts on cotton balls (72.1% sensitivity, 77.5% specificity) and cotton rolls (76.9% sensitivity, 76.5% specificity) ($\beta=0.01$, $z=0.01$, $P=0.99$), or on cotton balls and fecal matter (50% sensitivity, 90.4% specificity) ($\beta=-1.05$, $z=-1.79$, $P=0.07$ (Fig. 1). Post hoc pairwise comparisons also demonstrate that there is no

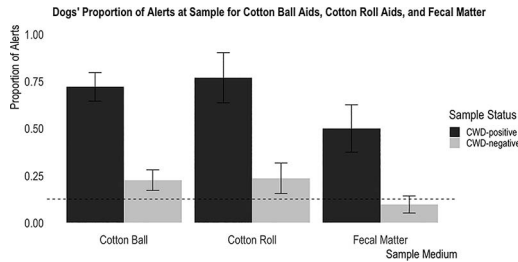


FIGURE 1. Proportion of trained alerts performed by detection dogs trained to detect chronic wasting disease (CWD) in fecal matter from white-tailed deer (*Odocoileus virginianus*) across three sample types (cotton ball aids, cotton roll aids, and fecal matter). Five dogs were trained at the Penn Vet Working Dog Center (Philadelphia, Pennsylvania, USA) using cotton balls incubated with fecal samples collected by the Wildlife Futures Program from CWD-positive and CWD-negative white-tailed deer. Bars represent the proportion of trials in which dogs performed their trained final alert at CWD-positive (black bars) and CWD-negative (gray bars) samples, with error bars indicating standard error of the mean. Dogs were tested under double-blind conditions using an eight-port scent wheel to evaluate their generalization of the target odor to novel samples and different sample substrates. The dotted line represents chance, at 12.5%.

difference in overall alerts between cotton rolls and fecal matter ($\beta=1.13$, $z=2.09$, $P=0.09$).

There was no interaction between Sample Status and Sample Medium (fecal matter versus cotton ball: $\beta_1=0.01$, $z=-0.01$, $P=0.99$; cotton roll versus cotton ball: $\beta_1=0.17$, $z=-0.18$, $P=0.85$). There was no interaction between Sample Status and Sample Medium (fecal matter versus cotton ball: $\beta_1=0.01$, $z=-0.01$, $P=0.99$; cotton roll versus cotton ball: $\beta_1=0.17$, $z=-0.18$, $P=0.85$). Further, post hoc pairwise comparisons of Sample Medium showed no significant interaction when comparing cotton roll and fecal matter for CWD-positive samples ($\beta_1=1.22$, $z=1.43$, $P=0.71$) and CWD-negative samples ($\beta_1=1.05$, $z=1.59$, $P=0.61$).

DISCUSSION

Given the risk of environmental contamination while handling the fecal matter taken from CWD-positive cervid species, this study evaluated the feasibility of training naïve dogs

on the odor of profile of CWD through cotton training aids and then testing them on actual fecal matter. The results showed 1) that there was an odor difference discernable to dogs between the CWD-positive and CWD-negative incubated cotton ball and cotton roll aids; 2) that dogs were able to generalize to some degree between trained cotton ball aids and the target fecal matter; and 3) that the type of cotton substrate used did not significantly impact dogs' performance.

Dogs trained on cotton ball aids were able to extend their learned CWD odor category to a never-before-seen medium for CWD odor: fecal matter. Dogs did, however, show a decrement in performance between trained cotton aids and untrained fecal matter. In the inverse scenario reported by Mallikarjun et al. (2024), dogs initially trained on fecal matter had a similar sensitivity on cotton aids as the dogs initially trained on cotton aids (73.3% sensitivity, as compared to the cotton-trained dogs' 73% sensitivity) but a marked reduction in specificity (61.9% specificity on cotton aids, as compared to the cotton-trained dogs' 77% specificity). This corroborates the idea that the cotton aid represents the signature CWD-positive and CWD-negative odors found in fecal matter, but it is not identical.

Previous studies have demonstrated that although dogs can extend categories of odor from one learned medium (e.g., blood plasma) to another unlearned medium (e.g., nasal secretions), their performance often decreases when first exposed to the novel medium (Mallikarjun et al. 2023a). The extent to which dogs are able to classify samples as CWD positive or CWD negative can be considered a form of learning described as "concept formation," whereby the animal learns to identify the complex and variable patterns that are indicative of a disease (Edwards et al. 2017; Crawford et al. 2022). If a novel odor is markedly distinct from a dog's trained concept of CWD positive odor, they will be unlikely to recognize this novel odor as their target. Our results indicate that the dogs were able to detect a consistency in odor across cotton aids and actual fecal matter, but their accuracy was lower for the fecal matter (50%

sensitivity and 90% specificity) as compared to their test performance on novel incubated cotton aids (73% sensitivity and 77% specificity). This finding suggests that the fecal matter may have been on the fringes of their concept. As such, it may be beneficial to expose detection dogs in training to actual fecal matter prior to becoming operational to incorporate this odor profile in the dog's concept of their target odor. The number of exposures required for optimal learning and accuracy would be of interest to assess in future studies directly, to establish a training plan that is both effective and minimizes risks of environmental contamination.

The second research question addressed whether the type of aid used (cotton ball or roll) impacted the dogs' performances. The balls and rolls were both 100% cotton, but were in different forms, compressed different amounts, and purchased from different suppliers. Cotton rolls may be more useful in field training as they are they are denser and less likely to be disrupted by the wind in the environment. We found no significant difference in the dogs' performances when using cotton balls and rolls. Given that rolls were found to be equivalent to balls, rolls may be preferable for outdoor training.

The cotton aid itself is unlikely to spread the infectious prion protein that causes CWD. When incubated in the headspace above the fecal matter, cotton does not come into direct contact with the biohazardous substance. Cotton adsorbs VOCs from the headspace above the cotton, and the prion protein itself is not found within the headspace (Ellis et al. 2019). This suggests that the cotton aids would be safe to place directly in the search environment.

One limitation to consider is that the cotton aids were incubated with 2-g samples of fecal matter, and the dogs were tested using 2 g of fecal matter. In a real-world scenario, white-tailed deer fecal pile weight can vary based on water content and diet, and ranges from mean 19.8 g for dried pellets to 122.4 g for fresh fecal piles (McCullough 1982). Previous studies have shown that dogs do not generalize well across different quantities of explosives (Aviles-Rosa et al. 2021). It is also generally important

to keep training scenarios and stimuli as similar to the testing and operational scenario as possible, or dogs' performance may decrease (DeChant et al. 2020). It was not possible for this study to use more than 2 g of fecal matter for incubation, because of constraints surrounding the availability of CWD confirmed positive samples. An important future question is how accurate dogs are at detecting large quantities of fecal matter from CWD-positive deer after training on small quantities in both fecal matter and in training aid form. Although small quantities of fecal matter are more accessible and portable for handlers, dogs may fail to find large samples in the field if only trained on small quantities. Training aids are often the same size, but it may not be possible to acquire enough large-quantity samples for incubation with training aids. Identifying a balance between practical considerations and dogs' needs is crucial for future detection work to be successful.

The development and testing of CWD training aids is crucial for the future operationalization of working CWD detection dogs. Some potential use cases for CWD detection dogs include monitoring of disease management areas (DMAs) to assess disease prevalence, as well as searching boundaries of DMAs to track and monitor the spread of CWD to other areas. The benefit of detection dogs is that they can traverse large areas of land much faster than human teams, and they can provide immediate preliminary information about CWD status, which no other test can do. Training aids allow detection dog handlers to train their dogs in areas similar to the actual DMAs without fear of contaminating novel areas with CWD.

There are several wildlife management implications that can be derived from these results. First, detection dogs can be trained to find fecal matter from CWD-positive deer, which may directly aid in management efforts to identify areas infected with CWD. Second, our finding that incubated cotton was an effective training aid for CWD detection dogs means that infected fecal matter does not need to be placed in the environment, risking environmental contamination, for training purposes.

Additionally, our findings indicate future research needed in this area. For example, there is a need to quantify the VOC profile of CWD. Ellis et al. (2019) examined the difference in VOCs in fecal matter from CWD-positive and CWD-negative deer; however, further studies are required to assess the VOC profiles of CWD training aids and the actual fecal matter. Additionally, further identification of the VOCs in training aids may aid in identification of the key volatiles needed for CWD recognition. A second area of applied interest is to investigate how long the CWD odor lasts in training aids. If one type of aid is found to hold the CWD odor longer and more consistently than the other types, it may play a factor in selection of ideal sample medium. Further, understanding the duration that a training aid can be used is important when utilizing aids in the field.

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